



EY

TalentFitter

CV Analyzer

Agent

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The Problem

Recruitment teams spend too much time screening CVs manually.

Every open position can receive dozens or hundreds of CVs. Manually comparing each candidate with the job requirements is slow, repetitive and difficult to scale.

Why it matters

- High volume of CVs
- Time-consuming manual screening
- Risk of missing strong candidates
- Limited explainability in shortlisting
- Recruiters need faster, data-driven support

Our Solution

An AI Agent that matches CVs to job descriptions automatically.

The system reads CVs and job descriptions, compares their semantic meaning, ranks the best candidates and explains the recommendation using Generative AI.

What it does

- Reads CVs in PDF format
- Reads job descriptions
- Converts text into embeddings
- Calculates candidate-job similarity
- Ranks top candidates
- Generates an AI recommendation
- Removes human bias

Technical Pipeline

The agent transforms unstructured CVs and job descriptions into comparable semantic representations.

This allows us to rank candidates based on meaning, not only keyword overlap.

How it works

- PDF text extraction with pypdf
- Text converted into embeddings
- 384-dimensional vector representation
- Cosine similarity calculates match score
- Gemini generates recommendation
- Gradio provides the user interface

Implementation Explained

Text is not directly understandable by mathematics.

The all-MiniLM-L6-v2 model transforms each CV and job description into a numerical vector with 384 dimensions.

Why it matters

- Captures semantic meaning
- Compares concepts, not only keywords
- Similar profiles are closer in vector space
- Enables more accurate candidate-job matching
- Match % is based on vector similarity

Prototype Demo

We built an interactive Gradio interface where recruiters can select a job position and instantly receive a ranked shortlist of candidates.

The system also generates an AI recommendation explaining why the top candidate is a good match.

Demo flow

- Select a job description
- Analyze all CVs automatically
- Display Top candidates
- Show Match %
- Generate Gemini recommendation
- Support recruiter decision-making

Responsible AI & Limitations

This solution is designed to support recruiters, not replace them.

The final hiring decision must remain human-led, with AI used as a decision-support tool.

Key considerations

- Match % is not a final hiring decision
- Human validation is required
- CV and job description bias may affect results
- PDF quality can impact text extraction
- Candidate privacy must be protected
- Recommendations should be auditable

Main Challenges

Building the prototype required balancing technical accuracy, explainability and usability within a limited timeframe.

Business Impact

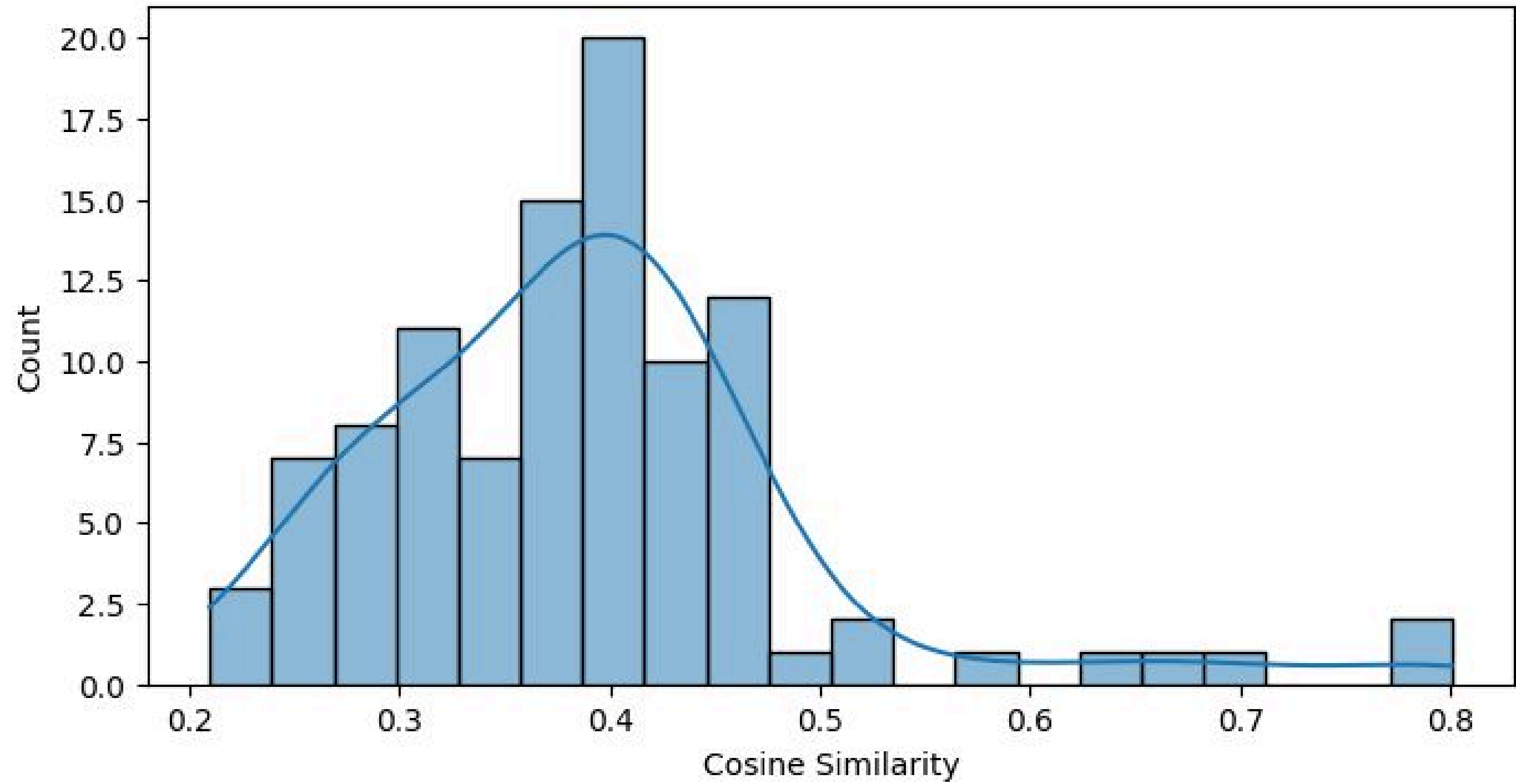
The solution acts as a recruitment copilot, helping teams move from manual screening to faster, data-supported shortlisting.

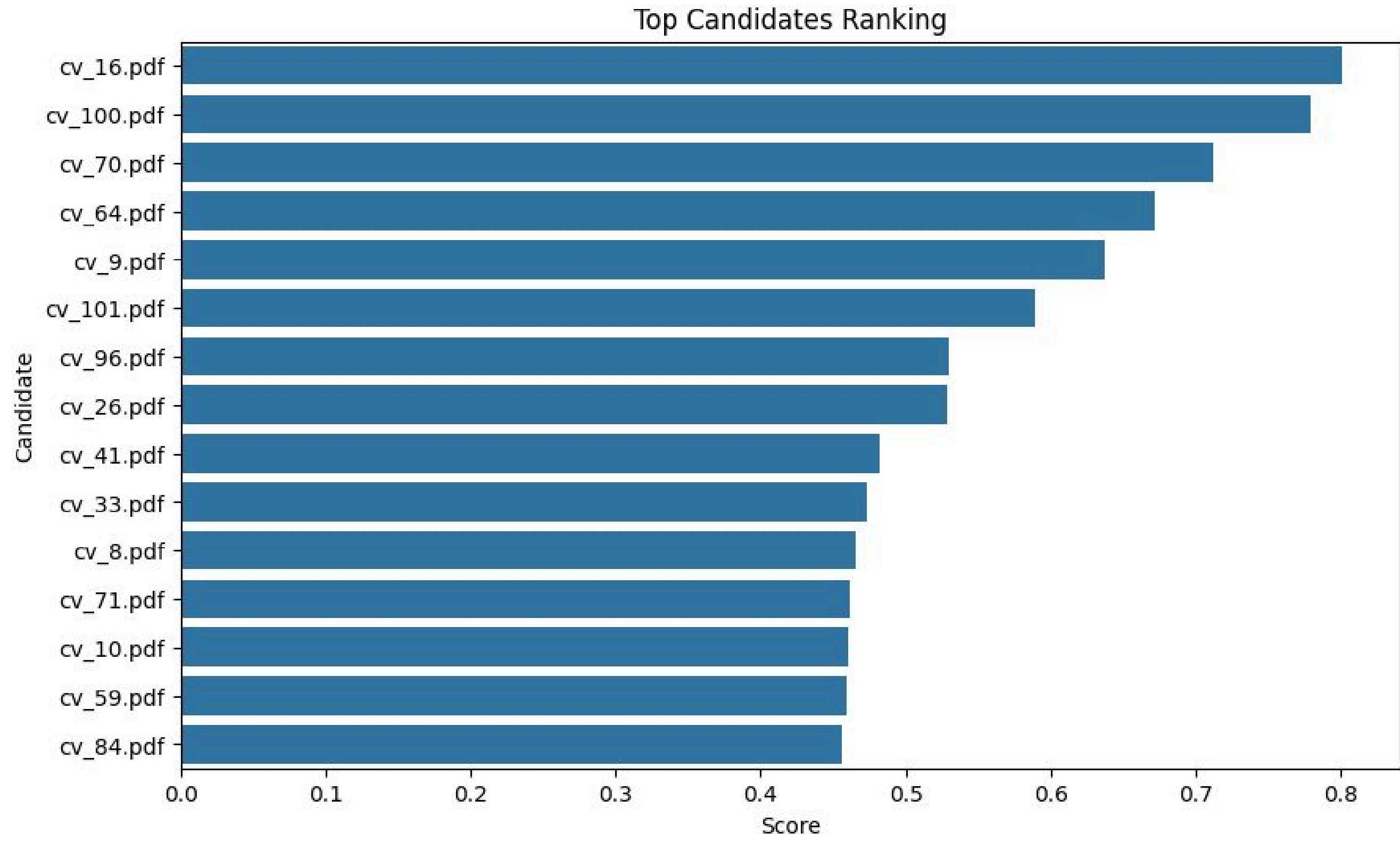
It does not replace recruiters — it helps them focus their time on the most promising candidates.

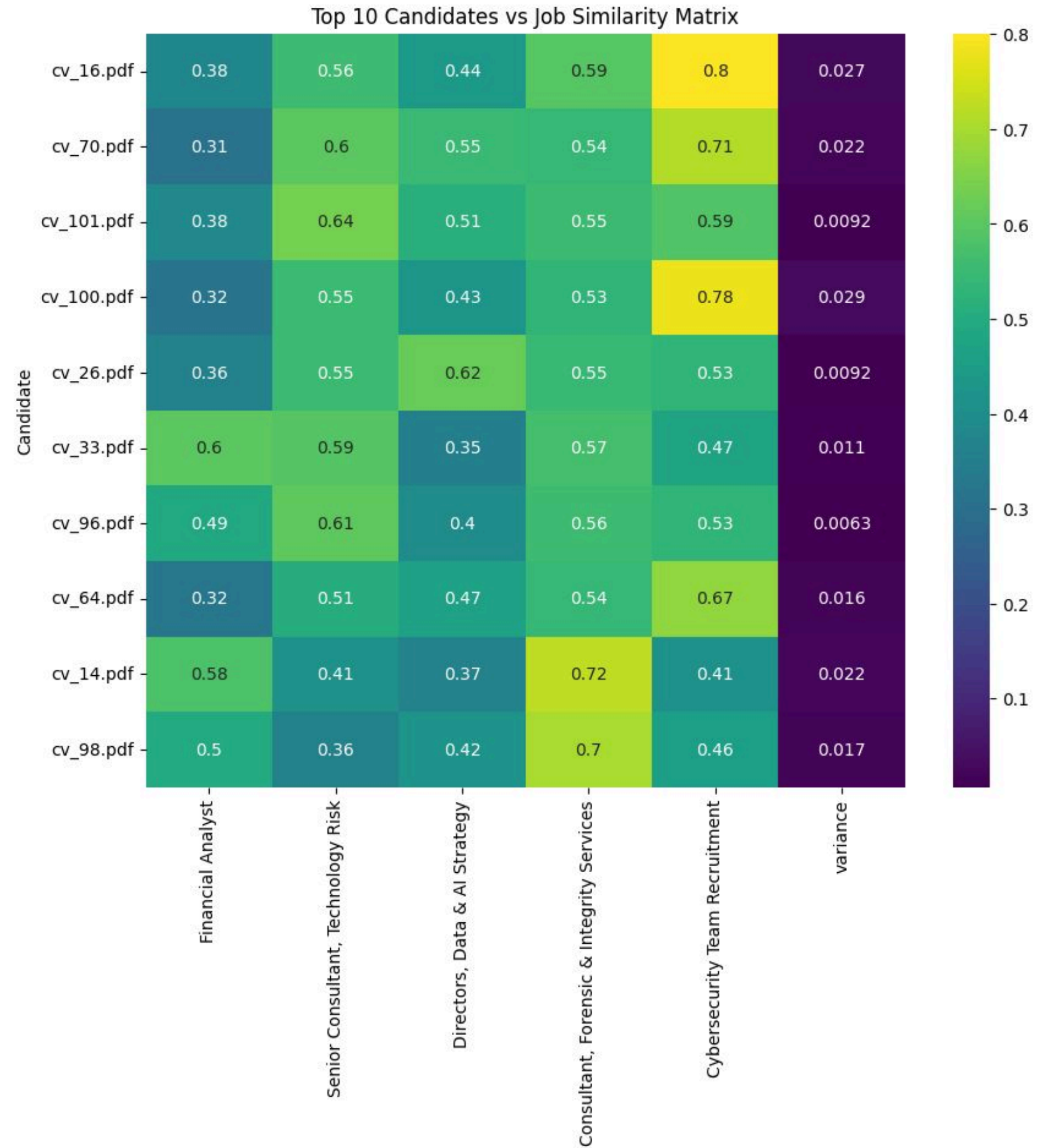
Value created

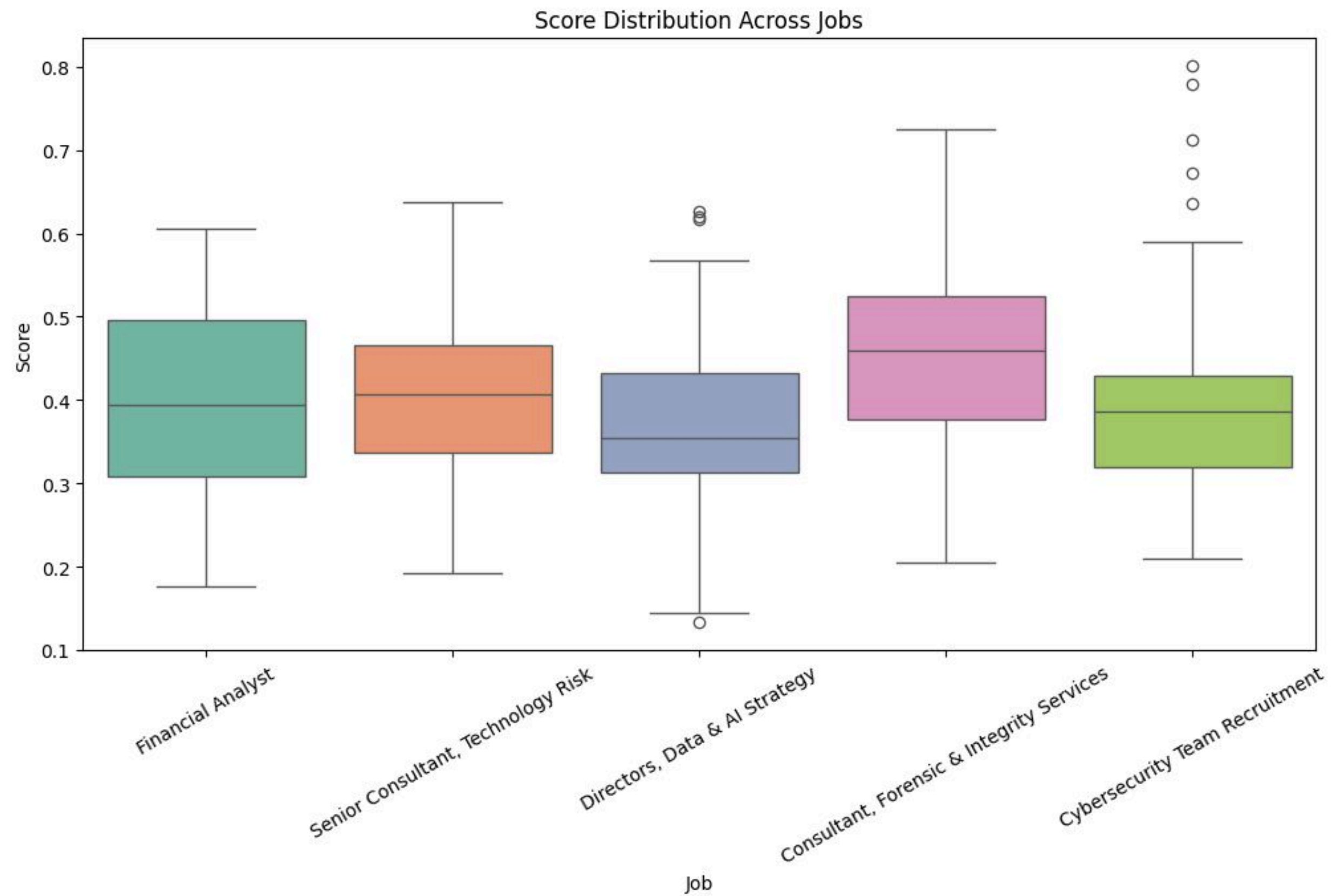
- Faster initial CV screening
- More consistent candidate shortlists
- Better prioritization of recruiter time
- Explainable recommendations
- Scalable across multiple job positions
- Human decision remains in control

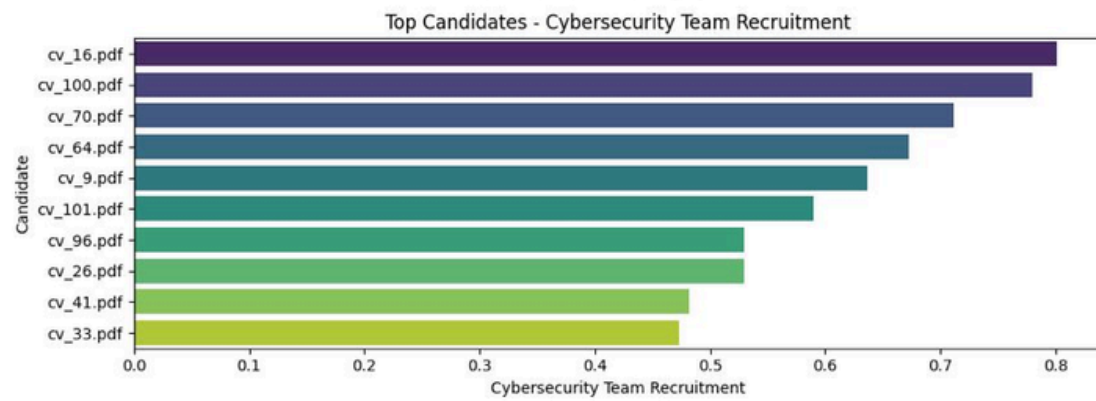
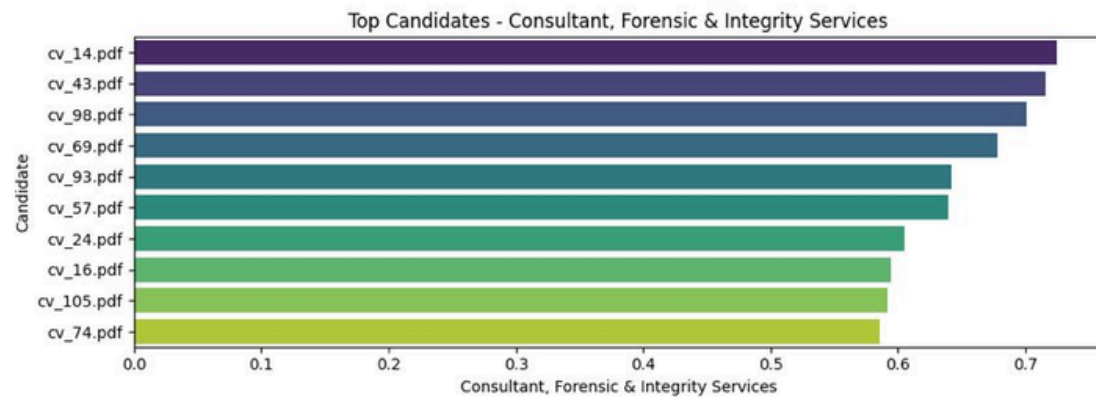
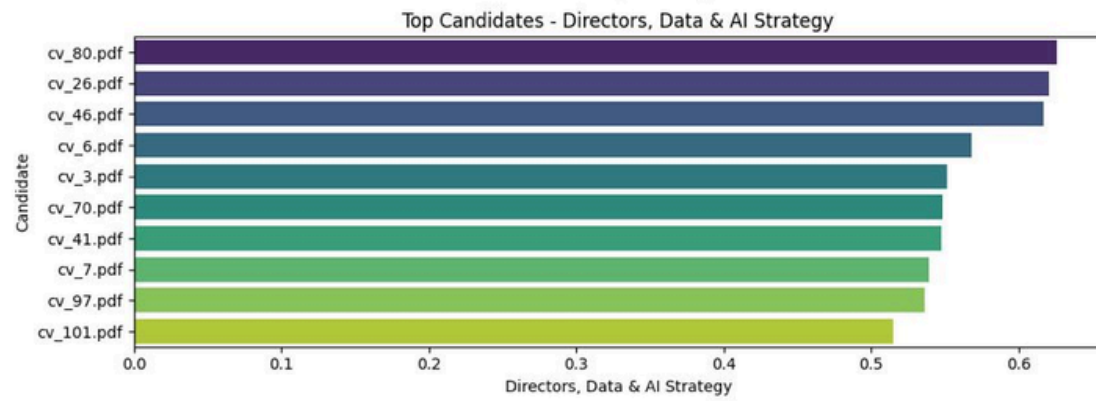
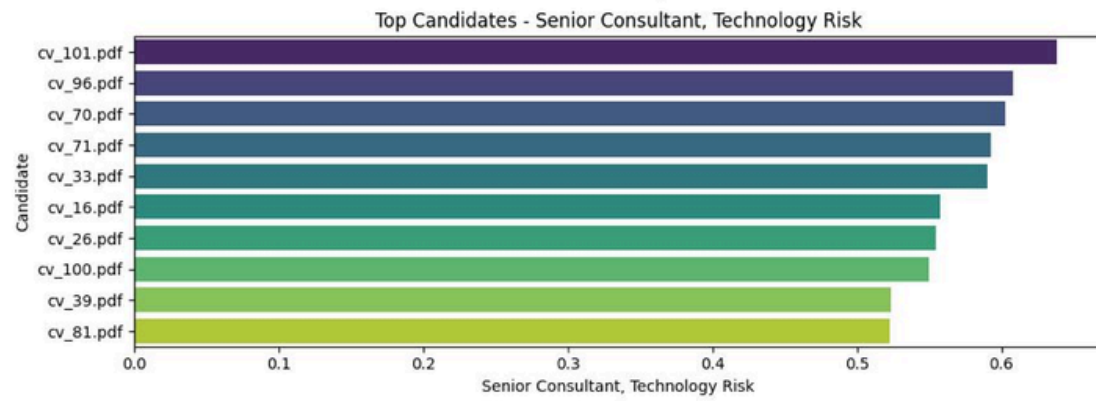
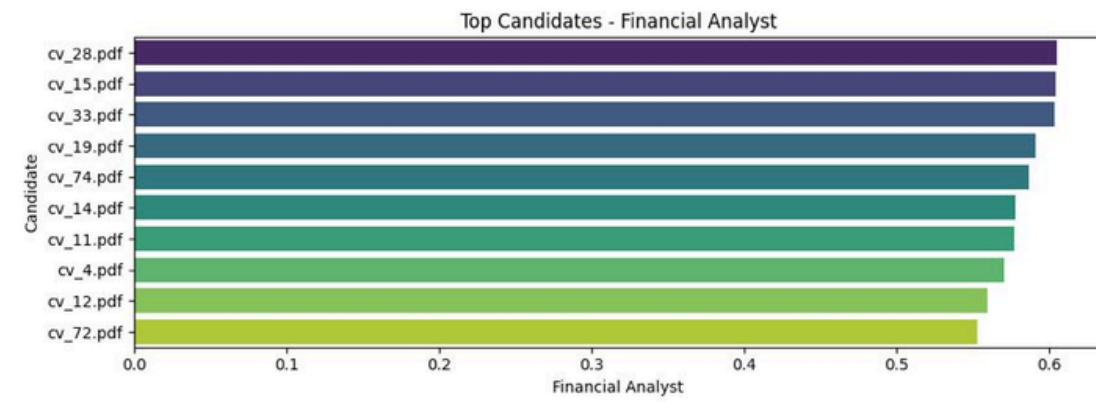
Distribution Match Scores



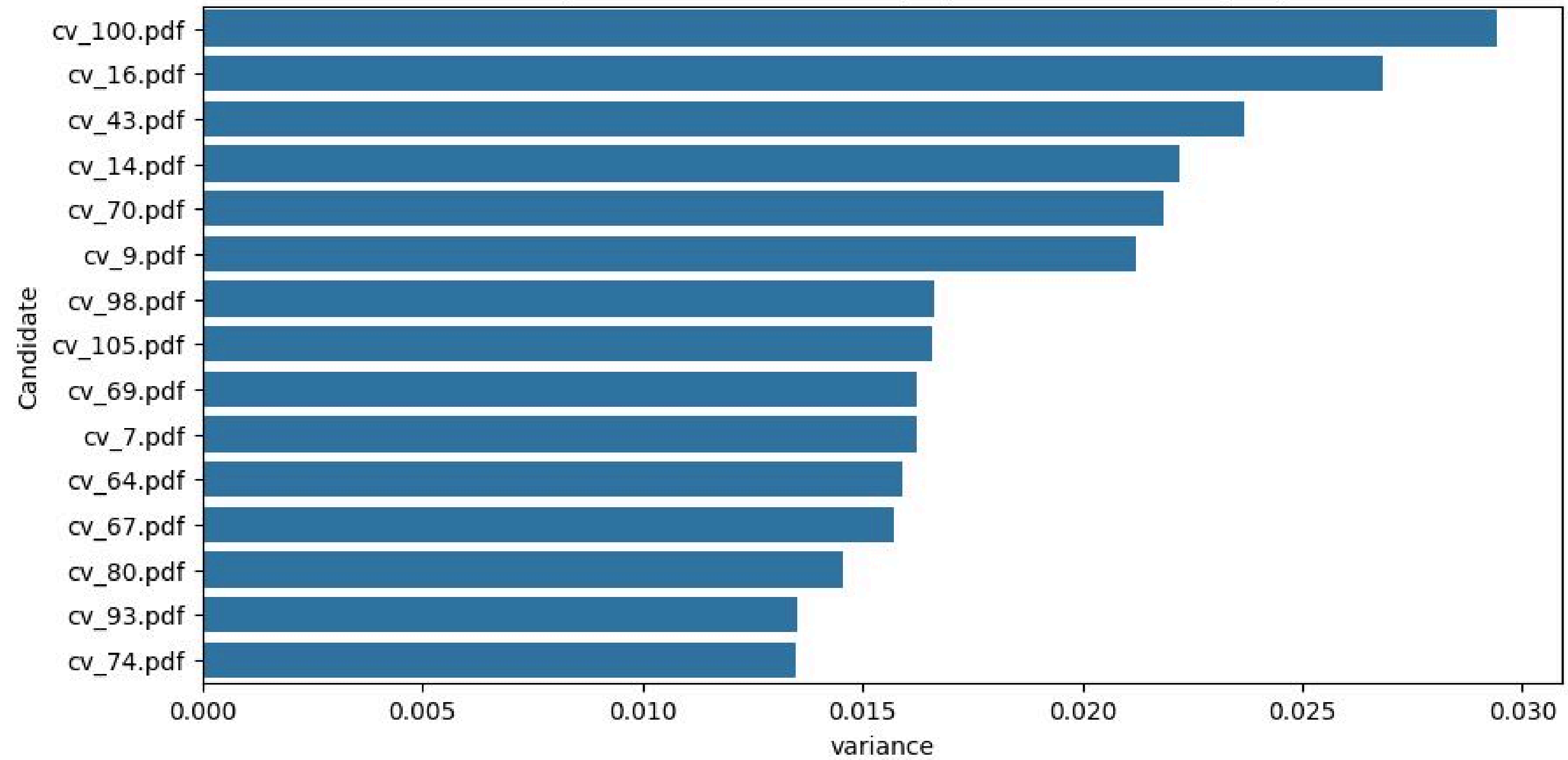


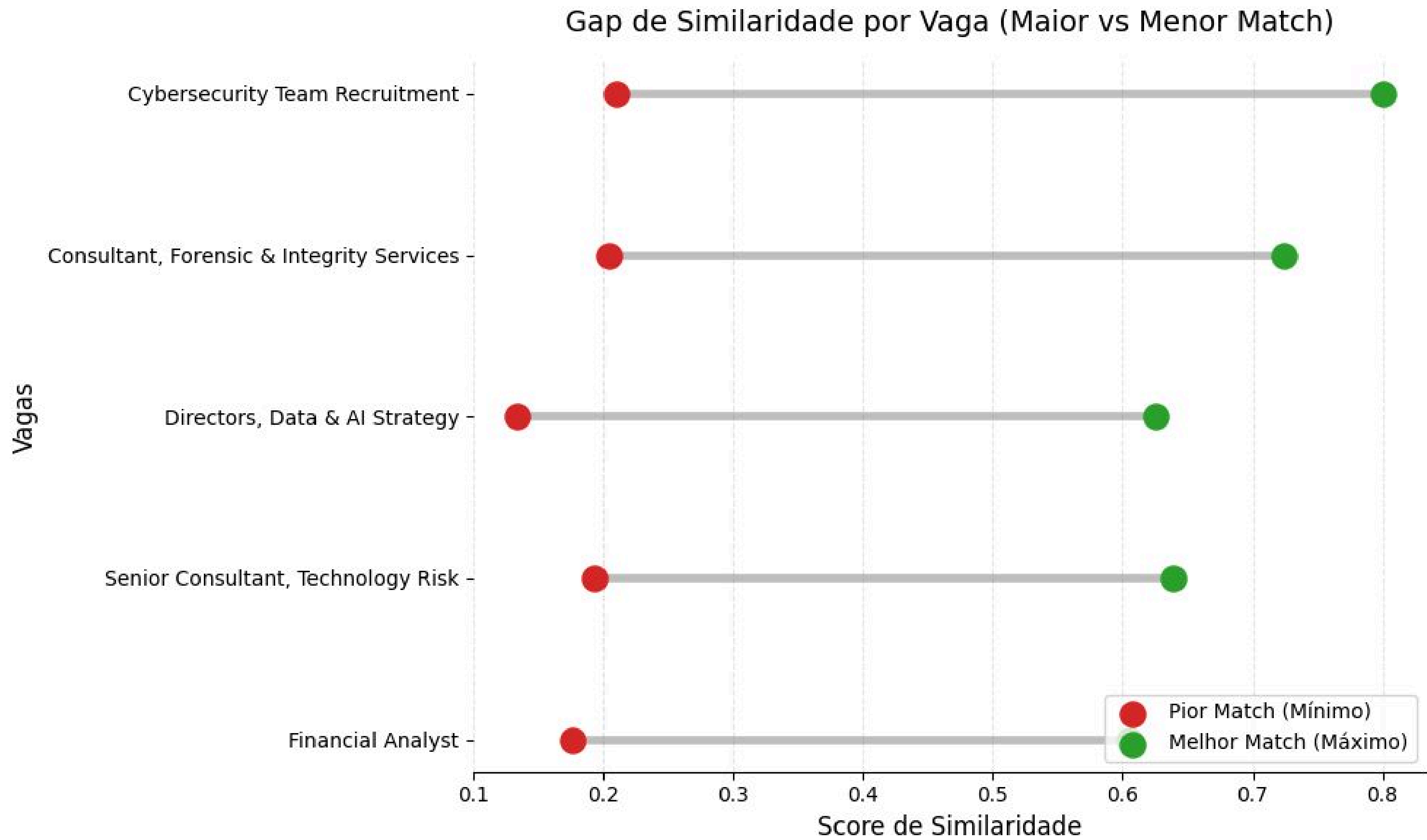






Most Specialized Candidates (High Variance Across Jobs)





Next Steps

The prototype can evolve from a proof of concept into a scalable recruitment intelligence tool.

The next phase would focus on deeper candidate insights, recruiter feedback and system integration.

Future improvements

- Extract skills and gaps automatically
- Add filters for seniority, location and role type
- Include recruiter feedback loop
- Validate recommendations with hiring outcomes
- Integrate with ATS / recruitment platforms
- Add audit logs for transparency